Causation, Probability, and Ledger Realism from Cost Geometry

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Abstract

We derive the nature of causation, the meaning of probability, and the structure of temporal reality from the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. Causation is identified with the ledger update: event C causally precedes event E if C occurs at an earlier tick and E's defect does not exceed C's. This causal relation is proved to be asymmetric, transitive, and irreversible. Probability is the fiber weight of a finite-resolution projection: the ledger is deterministic (unique J-minimizer at each tick), but any observer with resolution n partitions the state space into n fibers, and the natural measure on fibers constitutes probability. This makes probability relational (observer-dependent) and epistemic (about what can be known), not ontic (about what exists). Finally, we establish Ledger Realism as an alternative to both the block universe and pure presentism: the past (committed entries) is real, the present (current tick) is privileged, and the future (uncommitted entries) is genuinely open. Hume's problem of causation is dissolved: the causal connection is not a mysterious "necessary connexion" but a provable structural relation on the ledger.

1 Introduction

Three of the deepest questions in the philosophy of science concern causation, probability, and the nature of temporal reality.

Causation. Hume [**?**] argued that we never observe a "necessary connexion" between cause and effect—only constant conjunction. This has led to a spectrum of theories: regularity accounts [**?**], counterfactual analyses [**?**], interventionist frameworks [**?**], and process theories [**?**]. None derives causation from a more fundamental structure.

Probability. What do probability statements mean? Frequentism interprets probability as long-run relative frequency. Bayesianism treats it as degree of belief. Propensity theories locate probability in objective tendencies of physical systems. Each has known difficulties, and no consensus exists [**?**].

Temporal reality. The block-universe view holds that past, present, and future are equally real—time is a static four-dimensional manifold. Presentism holds that only the present exists. The growing-block theory adds the past as real but keeps the future open.

The debate connects to the problem of temporal passage: is "becoming" real, or is it an illusion? [**?**]

We show that Recognition Science (RS) [**?**] provides unified answers to all three questions from the same cost functional. Causation is ledger update. Probability is projection fiber weight. Temporal reality is Ledger Realism—a novel position that vindicates elements of both the block universe and presentism.

2 The Cost Functional and Ledger Dynamics

Definition 2.1 (Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. This is the unique normalized solution to the Recognition Composition Law. It satisfies $J(x) \geq 0$ with equality iff $x = 1$, $J(x) = J(1/x)$, and $J''(x) = x^{-3} > 0$.

Definition 2.2 (Ledger Snapshot)**.** A *ledger snapshot* is a pair $s = (t, d)$ where $t \in \mathbb{N}$ is the tick index and $d \geq 0$ is the total defect.

Definition 2.3 (Temporal Ordering)**.** Snapshot $s_1 = (t_1, d_1)$ is *before* $s_2 = (t_2, d_2)$ if $t_1 < t_2$.

Definition 2.4 (Recognition Step)**.** A *recognition step* is a transition $(t, d) \to (t{+}1, d')$ with $d' \leq d$.

The dynamics are deterministic: at each tick, the ledger transitions to the unique J-cost-minimizing configuration (guaranteed by the strict convexity of J).

3 Causation

3.1 Definition

Definition 3.1 (Causal Precedence)**.** Snapshot $s_1 = (t_1, d_1)$ *causally precedes* snapshot $s_2 = (t_2, d_2)$, written $s_1 \prec s_2$, if:

$$t_1 < t_2 \quad \text{and} \quad d_2 \leq d_1. \tag{1}$$

In words: s_1 occurs at an earlier tick, and s_2's defect does not exceed s_1's.

This definition captures two essential features of causation: temporal priority (the cause occurs before the effect) and defect transfer (the effect's defect is bounded by the cause's).

3.2 Structural Properties

Theorem 3.2 (Causal Asymmetry)**.** *Causal precedence is asymmetric: if $s_1 \prec s_2$, then $\neg(s_2 \prec s_1)$.*

Proof. $s_1 \prec s_2$ requires $t_1 < t_2$. For $s_2 \prec s_1$ we would need $t_2 < t_1$, which contradicts $t_1 < t_2$. □

Theorem 3.3 (Causal Transitivity)**.** *Causal precedence is transitive: if $s_1 \prec s_2$ and $s_2 \prec s_3$, then $s_1 \prec s_3$.*

Proof. $t_1 < t_2 < t_3$ gives $t_1 < t_3$. $d_3 \leq d_2 \leq d_1$ gives $d_3 \leq d_1$. □

Theorem 3.4 (Causal Irreversibility)**.** *If a recognition step strictly reduces defect ($d' < d$), the causal relation is irreversible: no subsequent step can restore the original defect.*

Proof. Any step from $(t{+}1, d')$ yields $(t{+}2, d'')$ with $d'' \leq d' < d$. □

Corollary 3.5 (Past Permanence)**.** *Past causal relations are permanent: once $s_1 \prec s_2$ is established, no future event can undo it.*

Theorem 3.6 (Causation–Arrow Agreement)**.** *In a defect-monotone sequence: the causal ordering $\prec$ agrees with the arrow of time (the direction of non-increasing defect). Formally, if the arrow points from s_1 to s_2, then $s_1 \prec s_2$, and conversely.*

Proof. The arrow of time from s_1 to s_2 means $t_1 < t_2$ and $d_2 \leq d_1$, which is exactly $s_1 \prec s_2$. □

3.3 Hume's Problem Dissolved

Theorem 3.7 (Hume Dissolved)**.** *Hume's skepticism about causation is dissolved. The "necessary connexion" between cause and effect is not a mysterious metaphysical relation but a provable structural feature of the ledger:*

(i) There exists a unique zero-defect state ($x = 1$).
(ii) For all $x > 0$ with $x \neq 1$, defect is strictly positive: $J(x) > 0$.
(iii) Defect monotonically decreases along the tick sequence.

The "necessary connexion" is the mathematical necessity of defect non-increase: the cost functional forces *the causal relation.*

Proof. (i) $J(1) = 0$. (ii) $J(x) > 0$ for $x \neq 1$ by AM–GM. (iii) Recognition steps do not increase defect (Definition **??**). These three facts jointly establish that the causal ordering is not contingent or conventional but is structurally forced by the cost functional. □

Remark 3.8. Hume could not find the "necessary connexion" because he looked for it in the wrong place: in the sensory experience of constant conjunction. In RS, the connection is in the mathematics of the cost landscape. It is "necessary" in precisely the sense that a strictly convex function *necessarily* has a unique minimizer.

4 Probability

4.1 The Projection Framework

Definition 4.1 (Observer)**.** An *observer* is specified by a positive integer n (the *resolution*): the number of distinguishable output states.

Definition 4.2 (Projection Map)**.** Given an observer with resolution n, the *projection map* $\pi_n : \mathbb{R} \to \{0, 1, \ldots, n{-}1\}$ assigns each real number to an output label.

Definition 4.3 (Fiber)**.** The *fiber* of outcome k is:

$$F_k := \pi_n^{-1}(k) = \{x \in \mathbb{R} : \pi_n(x) = k\}. \tag{2}$$

Theorem 4.4 (Fiber Partition)**.** *The fibers $\{F_0, \ldots, F_{n-1}\}$ partition the state space:*

(i) Every state belongs to exactly one fiber.
(ii) Each fiber is non-empty.
(iii) The fibers are exhaustive: $\bigcup_k F_k = \mathbb{R}$.

Proof. (i) and (iii): π_n is a total function. (ii) For each k, there exists at least one x with $\pi_n(x) = k$. □

Theorem 4.5 (Projection is Lossy)**.** *For every observer with resolution $n \geq 1$, the projection map is many-to-one: there exist distinct $x \neq y$ with $\pi_n(x) = \pi_n(y)$.*

Proof. The real line is uncountable; the output set has n elements. By the pigeonhole principle, each fiber contains uncountably many elements. □

4.2 Probability as Fiber Weight

Definition 4.6 (Observer Probability)**.** Given an observer with resolution n and a symmetric uniform partition of the state space, the *observer probability* of outcome k is:

$$P(k) = \frac{\mu(F_k)}{\sum_{j=0}^{n-1} \mu(F_j)}, \tag{3}$$

where μ is the natural measure on the state space. For the J-cost landscape with $x > 0$, the natural measure on the positive reals is the Haar measure $\mu_{\mathrm{H}}(A) = \int_A \frac{dx}{x}$ (the log-flat measure), which assigns equal weight to each φ-rung interval. For a uniform φ-rung partition into n equal-width log-intervals, $\mu_{\mathrm{H}}(F_k) = \ln \varphi$ for all k, giving $P(k) = 1/n$.

Remark 4.7. The choice of Haar measure (log-flat) is natural for the RS framework because the φ-ladder is a multiplicative group action: $x \mapsto \varphi x$ maps rungs to rungs. The unique measure invariant under this action is the log-flat measure. Any observer who partitions the φ-ladder into equal-width log-intervals therefore assigns uniform probability to each interval, and $P(k) = 1/n$ follows. Non-uniform partitions correspond to observers with structure-dependent bias and lead to non-uniform probability distributions—which is the RS derivation of non-uniform priors in Bayesian probability.

Theorem 4.8 (Probability Axioms Satisfied)**.** *Observer probabilities satisfy the Kolmogorov axioms:*
(i) $P(k) \geq 0$ for all k (non-negativity).
(ii) $\sum_{k=0}^{n-1} P(k) = 1$ (normalization).
(iii) For disjoint outcomes $A, B \subseteq \{0, \ldots, n-1\}$: $P(A \cup B) = P(A) + P(B)$ (σ-additivity, finite form).

Proof. (i) $P(k) = 1/n > 0$. (ii) $\sum_k 1/n = n \cdot (1/n) = 1$. (iii) For disjoint finite sets, $P(A \cup B) = |A \cup B|/n = (|A| + |B|)/n = |A|/n + |B|/n = P(A) + P(B)$, since $|A \cup B| = |A| + |B|$ for disjoint sets. □

4.3 The Meaning of Probability

Theorem 4.9 (Probability is Epistemic)**.** *(i) The ledger is deterministic: there exists a unique zero-defect state, and the dynamics have a unique J-minimizing next state.*
(ii) Randomness arises from the lossy projection: multiple distinct deterministic states map to the same observation.

(iii) Therefore, probability is epistemic (about what an observer can know), not ontic (about what exists).

Proof. (i) $J(x) = 0$ if and only if $x = 1$ (by AM–GM), so there is a unique zero-defect state. The cost functional is strictly convex ($J''(x) = x^{-3} > 0$), so the J-minimizing next state is unique at each tick. The ledger dynamics are thus deterministic.

(ii) Theorem **??** establishes that the projection map is many-to-one: the fiber $F_k = \pi_n^{-1}(k)$ contains uncountably many distinct ledger states. An observer recording outcome k cannot determine which element of F_k is the actual ledger state—this is irreducible uncertainty about the underlying deterministic state.

(iii) Since the underlying ledger state is determinate (by (i)) and the probability distribution arises entirely from the observer's finite resolution (by (ii)), probability statements say nothing about the ledger itself—they describe the observer's partition of the state space. This is the definition of epistemic probability. □

Remark 4.10 (Reconciling Frequentism, Bayesianism, and Propensity). RS vindicates elements of all three interpretations:

- **Frequentism**: the fiber structure is objective, so long-run relative frequencies are objective features of the projection.
- **Bayesianism**: probability depends on the observer's resolution n—a higher-resolution observer has finer fibers and different probabilities. Probability is thus relational to the observer's information state.
- **Propensity**: the J-cost landscape is objective and determines which configurations are "favored" (low cost). This provides an objective "tendency" structure.

The three interpretations are not competitors but complementary aspects of the same underlying structure.

Theorem 4.11 (Higher Resolution Refines Probability)**.** *If observer 2 has strictly higher resolution than observer 1 ($n_2 > n_1$), then every outcome of observer 1 can be further distinguished by observer 2: there exists a surjection $\rho : \{0, \ldots, n_2-1\} \to \{0, \ldots, n_1-1\}$ such that the partition by π_{n_2} refines the partition by π_{n_1}. Consequently, $P_{n_2}(k') \leq P_{n_1}(\rho(k'))$ for each outcome k', with equality only when $n_2 = n_1$.*

Proof. Any partition into $n_2 > n_1$ equal-log intervals is a refinement of the partition into n_1 intervals. The surjection ρ assigns each fine interval to the coarse interval containing it. The probability $P_{n_2}(k') = 1/n_2 < 1/n_1 = P_{n_1}(\rho(k'))$ strictly decreases at each refinement step. □

Remark 4.12. In the limit $n \to \infty$, the observer's resolution approaches the full ledger state, probability concentrates on the unique deterministic outcome ($P \to \delta_{x_{\min}}$), and "randomness" vanishes. This is why quantum randomness appears irreducible only at finite resolution: it is a feature of the projection, not of the reality projected.

5 Ledger Realism

5.1 The Problem of Temporal Reality

The block-universe view holds that past, present, and future are equally real. Presentism holds that only the present exists. Both face difficulties: the block universe struggles with

temporal passage and the apparent openness of the future; presentism struggles with cross-temporal relations and the truth-conditions of tensed statements.

RS offers a third position: *Ledger Realism.*

5.2 Past, Present, Future

Definition 5.1 (Past, Present, Future)**.** Given a sequence of snapshots $\{s_n\}$ and a designated "now" index k:

(a) The **past** is $\{s_n : n < k\}$—committed, permanent ledger entries.
(b) The **present** is s_k—the current tick, ontologically privileged.
(c) The **future** is $\{s_n : n > k\}$—uncommitted entries.

Theorem 5.2 (Past Permanence)**.** *Past entries are real and permanent: for every $n < k$, snapshot s_n is a committed entry in the ledger and cannot be altered by any future recognition step.*

Proof. Recognition steps advance the tick ($t \to t{+}1$) and do not modify earlier entries. Once committed at tick n, the snapshot s_n is fixed. □

Theorem 5.3 (Present is Privileged)**.** *The present snapshot s_k is the unique snapshot at the current tick:* $\mathrm{present}(\{s_n\}, k) = s_k$.

Theorem 5.4 (Time Divides into Three)**.** *The present divides the ledger into exactly two non-overlapping regions:*

(i) $s \in \mathrm{past} \Leftrightarrow \exists\, n < k$ *with* $s_n = s$.
(ii) $s \in \mathrm{future} \Leftrightarrow \exists\, n > k$ *with* $s_n = s$.

No snapshot belongs to both past and future.

Theorem 5.5 (Temporal Quantization)**.** *Time is discrete: for any two snapshots with s_1 before s_2,*

$$t_2 - t_1 \geq 1. \tag{4}$$

The minimal temporal resolution is one tick.

Theorem 5.6 (8-Tick Specious Present)**.** *The epoch length is $8 = 2^D = 2^3$ ticks (forced by the dimension-forcing theorem). One complete recognition cycle spans 8 ticks, providing a natural "specious present"—the minimal window of simultaneous information processing.*

Remark 5.7. The connection between the 8-tick fundamental epoch and the psychological specious present (the experienced "now", empirically $\sim$ 2–3 s) requires identifying a conversion factor between recognition ticks and clock seconds. Using the RS tick duration $\tau_0 = \hbar/E_{\mathrm{coh}} \approx 10^{-13}$ s, eight ticks span $\sim 8 \times 10^{-13}$ s—far shorter than the psychological specious present. The psychological specious present is likely the result of many nested 8-tick epochs processed hierarchically by the neural ledger, not the fundamental 8-tick cycle itself. This is an open problem in the application of RS to neuroscience.

5.3 Ledger Realism as a Third Way

Theorem 5.8 (Block Universe Partially Vindicated)**.** *The block-universe view is correct insofar as the past is real and permanent (Theorem* **??***). Past entries exist and have definite values, just as the block universe claims.*

Theorem 5.9 (Presentism Partially Vindicated)**.** *Presentism is correct insofar as the present is ontologically privileged (Theorem* **??***). The current tick is not merely a coordinate label but a structurally distinguished point in the ledger.*

Remark 5.10 (Ledger Realism)*.* Ledger Realism is a growing-block theory with a twist: the past is real (block-universe element), the present is privileged (presentist element), and the future is uncommitted. It differs from the standard growing-block theory [**?**] in that the "growth" is not continuous addition of temporal slices but discrete tick-by-tick ledger commits. The irreversibility of recognition steps ensures that the past cannot be altered.

6 Discussion

6.1 Comparison with Existing Theories

	Hume	**Lewis**	**Woodward**	**RS**
Causation	Conjunction	Counterfactual	Intervention	Ledger update
Necessity	Denied	Modal	Structural	Proved
Asymmetry	Conventional	World-similarity	Intervention dir.	Tick ordering
Time	Subjective	B-theory	Neutral	Ledger Realism

Lewis's counterfactual analysis. Lewis [**?**] analyzes "C causes E" as: in the closest possible world where C does not occur, E does not occur either. RS provides a simpler account: C causes E because C's ledger entry constrains E's. No appeal to possible worlds is needed.

Woodward's interventionism. Woodward [**?**] analyzes causation via interventions: C causes E if intervening on C would change E. RS is compatible: an "intervention" is a modification of a ledger entry, and the causal chain propagates through subsequent tick updates. But RS is more fundamental: it derives the causal structure rather than taking interventions as primitive.

Pearl's structural causal models. Pearl [**?**] formalizes causation via directed acyclic graphs (DAGs) and structural equations. In RS, the ledger's tick-ordered entries are a DAG by construction (Theorems **??** and **??**), and the structural equation is the J-cost minimization rule. RS thus provides a physical grounding for Pearl's formal machinery: the DAG arises from tick ordering, and the "structural equations" are the RS cost dynamics.

6.2 Falsifiable Predictions

1. **Causal irreversibility.** RS predicts that no physical process at the fundamental level can reverse a causal relation (restore a prior defect while advancing the tick counter). An observation of fundamental causal reversal would refute the framework.

2. **Discrete causation.** Causal connections have a minimal temporal resolution of one tick. If causation is found to operate continuously at arbitrarily fine time scales (below the Planck time), the discrete ontology is falsified.

3. **Probability as projection.** The Born rule of quantum mechanics is recovered in the limit of large fiber count n. At very high resolution (approaching the fundamental tick scale), the fiber structure is discrete and the Born rule probability distribution may show small corrections of order $1/n$. Whether such deviations are observable at accessible energy scales requires a quantitative connection between n and the experimental probe scale—an open problem. This prediction is speculative and the direction of the correction depends on the specific choice of partition.

4. **Past permanence.** RS predicts that past events cannot be altered. Any demonstration of backward causation (genuine retrocausation, not merely retrodiction) would refute the framework.

7 Conclusion

Three foundational questions in philosophy of science—the nature of causation, the meaning of probability, and the structure of temporal reality—receive unified answers from the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$:

1. **Causation** is the ledger update relation $\prec$: asymmetric, transitive, irreversible, and grounded in defect monotonicity. Hume's problem is dissolved: the "necessary connexion" is the mathematical necessity of cost minimization.
2. **Probability** is the fiber weight of a finite-resolution projection. It is epistemic (about observation, not reality), relational (dependent on observer resolution), and objective (the fiber structure is determined by the projection map). Frequentism, Bayesianism, and propensity are all partially vindicated.
3. **Temporal reality** is Ledger Realism: the past is real and permanent, the present is privileged, the future is uncommitted. This vindicates elements of both the block universe and presentism while avoiding the difficulties of each.

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